Macao Tourism Recovery

Visitor composition, daily arrivals and non-gaming expenditure

CISC7204 Data Science and Data Visualization
Project Research Proposal
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AI acknowledgment: OpenAI Codex assisted with source checks, course mapping, code, charts and editing. The accompanying data and code support review of the results. Recorded values come from the cited official sources; generated text does not replace observations (OpenAI, 2026; University of Macau, 2026b).

Team details, collaboration arrangements and the four-week schedule are deferred. This version covers the research content.

Research overview

We examine whether the recovery in Macao visitor arrivals was accompanied by similar changes in daily distributions, visitor composition and non-gaming expenditure. Extending the tourism topic in Assignment 2, we combine 3,287 observed dates from 2017 to 2025, 55,879 date-by-source records, and 16 expenditure quarters covering 2019 and 2023-2025. Two notebooks implement data checks, descriptive and composition analyses, calendar regression and the quarterly spending comparison.

In 2025, total arrivals were 1.68% above 2019, but the daily median was slightly lower, P90 and IQR were higher, and the overnight share fell by 5.99 percentage points. Ridge was selected in 2024 validation, while OLS with a trend had lower 2025 test error. Official nominal spending per visit was about 23.00% above 2019 but 7.28% below 2024. Recovery therefore depends on the measure, distribution, composition and comparison period.

The project uses taught methods: data wrangling, descriptive statistics, empirical cumulative distributions, stratified comparisons, multiple linear regression, Ridge and model evaluation (University of Macau, 2026c, 2026g, 2026h, 2026k, 2026l, 2026m). Its depth comes from checking measurement definitions, joining data at compatible levels, comparing models honestly and explaining errors within the evidence.

Purpose and learning outcomes

We extend Assignment 2 from aggregate tourism spending to the joint study of arrivals, composition and expenditure. The intended learning outcomes are to check official data and observation units with Pandas; explain differences behind totals with distributions and stratified plots; and compare OLS and Ridge with clear training, validation and test boundaries. Tourism knowledge helps distinguish visits from people, overnight visitors from hotel guests, and nominal spending from real purchasing power. The completed analyses provide evidence of these skills.

Research questions

ID	Question	Evidence
RQ1	How did total arrivals, overnight/same-day composition and source composition change between 2019 and 2025?	Annual arrivals, visitor-type shares, source shares and absolute changes

ID	Question	Evidence
RQ2	How did the typical level, spread and upper quantiles of daily arrivals change, and did changes vary by month and weekday?	Mean, median, IQR, P90, ECDF and calendar strata
RQ3	How did calendar models selected using 2024 validation perform in 2025, and how did trend and Ridge affect errors?	Baseline, OLS and Ridge; chronological validation, test metrics and high-arrival errors
RQ4	Did arrivals, nominal non-gaming expenditure and official spending per visit recover together under same-quarter and annual comparisons?	Same-quarter 2019 indices; 2025 versus 2019 and 2024 annual comparisons; type shares and means

RQ1 and RQ2 describe observed historical changes. RQ3 evaluates estimation errors in a later period. RQ4 extends the original spending question and compares overall and visitor-type means. Associations and arithmetic contributions to change are not interpreted as causal effects of policies, visitor types or source markets.

Data sources and scale

Dataset	Coverage and size	Use
DSEC tourism statistics database	1 January 2017 to 31 December 2025; 3,287 dates	Daily totals and visitor types
Source breakdown from the same database	17 mutually exclusive document-issue groups; 55,879 records	Source composition and grouped comparisons
DSEC visitor expenditure survey	16 quarters in 2019 and 2023-2025	Total and type-specific spending, including spending per visit

Two datasets are available through https://www.dsec.gov.mo/TourismDBWeb/ and https://www.dsec.gov.mo/en-US/Statistic?id=404. They provide administrative arrival counts and survey-based expenditure estimates, respectively (Statistics and Census Service, n.d.-a, n.d.-b). Daily breakdowns are available from 2017 (Statistics and Census Service, 2017). The full source archive retains official responses, request details, quarterly publications and extraction locations; the analysis package supplies the fixed input tables and source URLs.

The 55,879 source-group records share 3,287 dates and are not independent time observations. The periods 2017-2019 and 2023-2025 contain 2,191 dates and 37,247 source-group records. The 1,096 dates in 2020-2022 remain in the full history and are marked separately. Regression uses the 1,096 dates in 2023-2025.

Arrivals count visits, so one person may appear more than once. Source groups follow the place of travel-document issue, not nationality or residence. Overnight arrivals are not hotel guests or room nights. Expenditure is published from a quota survey and extrapolation; respondent records, survey weights and usable variance estimates are unavailable. All expenditure is in current-price Macao patacas (MOP) and excludes gaming.

Data preparation and quality checks

Source data remain unchanged. Daily total and visitor-type tables use date as the unique key; source records use date and source code. We check continuous dates, duplicate keys, non-negative counts, missing values and component sums, then use Pandas groupby, merge and pivot. Source codes are categories, not numerical distances.

All 3,287 dates are present. Each date has 17 source groups whose sum equals total arrivals. The same-day count is missing on 16 September 2018; its raw missing value is retained, alongside a separately flagged derived value of zero from total minus overnight arrivals. Eight other dates have one-arrival type-sum discrepancies, which remain unadjusted. All nine flagged dates precede 2023 and do not affect the main regression sample.

We aggregate 1,461 daily records from 2019 and 2023-2025 into complete calendar quarters, then join the 16 expenditure observations one-to-one by year and quarter. Every quarter matches and contains all expected dates. The daily rows supply count denominators; the spending unit remains the city-quarter. Extraction distinguishes single-quarter, cumulative and annual columns, checks older 2019 layouts and changed 2025 column order, and records locations for 96 quarterly indicators.

Official spending per visit and values derived as published expenditure divided by arrivals are stored separately. Across 48 checks, the maximum absolute relative differences are 0.0211% overall, 0.0148% for overnight visitors and 0.0797% for same-day visitors. These small differences support matching units and periods, not an unbiased survey. Quarterly spending sums are MOP 1 million above the official annual value in 2019 and MOP 1 million below it in 2024. Both versions remain available.

Experiment 1 Arrival distributions and composition

For the fixed periods 2017-2019, 2020-2022 and 2023-2025, we calculate observed days, means, medians, IQR and P90, and compare the 2019 and 2025 empirical cumulative distribution functions (ECDFs). Means and quantiles distinguish total recovery from changes in typical and upper-tail days. A high-arrival threshold is a statistical definition, not an official capacity or congestion standard.

Comparisons are stratified by month, weekday, and their combination. Monthly plots and the year-by-month heatmap use daily means; weekday plots use medians. All genuine high and low values remain. IQR describes spread and is not a deletion rule. These strata do not align lunar holidays, weather or events, remove all confounding, or justify default confidence bands that assume independent days.

The annual overnight share is annual overnight arrivals divided by annual total arrivals, not the unweighted mean of daily shares. Each source group is compared by counts, shares and absolute change from 2019 to 2025. Group changes must add to total change. Counts and percentage points help interpret opposing changes when the net change is small.

Totals, components, ratios and denominators are mathematically related, so overall and stratified results are reported together. We do not calculate correlation coefficients, test a Simpson reversal or identify behavioural mechanisms. The taught ideas of stratification and confounding guide the interpretation (University of Macau, 2026g).

Experiment 2 Calendar regression

The target is the raw daily total. Candidates are a training-period weekday-mean baseline, OLS with month and weekday, OLS adding a linear date trend, and Ridge with the same trend features. Month and weekday use one-hot encoding with a reference category dropped. Trend is days since 1 January 2023.

There are no interactions, holiday, weather or lag features, or target transformations. Same-day source and visitor-type counts are excluded because they sum to the target.

Stage	Fixed design
Initial training	All 365 days in 2023
Validation	Each 2024 quarter is one validation window; training uses only earlier dates in 2023-2024
Ridge candidates	Alpha = 0.01, 0.1, 1, 10, 100, 1000; selected by mean fold MSE
Final fitting	Refit on all 731 days in 2023-2024 after selection
Final test	All 365 days in 2025; no changes to features or parameters based on test results

Encoding and scaling are fitted inside each training fold through a Pipeline. StandardScaler affects only the continuous trend; month and weekday dummies stay at 0/1. GridSearchCV uses explicit time indices. Alpha and the candidate model are chosen by the equally weighted mean of four 2024 fold MSEs; validation RMSE is the square root of that mean MSE. The expanding windows apply taught validation principles to ordered data; they are not a new specialist time-series model or a claimed copy of a course example.

Primary evaluation uses MSE, RMSE and R^2; selection uses 2024 validation MSE only. MAE is an auxiliary description of mean absolute residuals, applying the taught ideas of means and residuals, rather than a separately claimed Module 05 topic. RMSE gives more weight to large errors. R^2 uses the observed test-period mean as an evaluation reference, can be negative, and is not prediction accuracy. Residuals equal actual minus estimated arrivals, so positive values mean underestimation. Errors are examined by month, weekday and observed arrivals above the 2023-2024 training P90.

The 2025 data do not enter fitting, alpha search or selection scores, but they were viewed in the earlier descriptive analysis. This is therefore a historical holdout, not a fully blind test. Alpha search and candidate comparison share the 2024 folds; there is no nested validation. Publication latency is unknown. We do not establish real-time or future performance, test residual autocorrelation, or test coefficient significance.

Experiment 3 Linking arrivals and expenditure

The 16 quarters extend the spending question in Assignment 2. We compare arrivals, overnight shares, non-gaming expenditure and spending per visit by visitor type. Each recovery index uses the corresponding 2019 quarter as 100. Counts, money, per-visit amounts and shares retain clear units; raw values with incompatible units are not placed on the same axis.

Official annual spending per visit comes from annual publication tables, not an average of four quarterly values. Official and derived values remain separate. Visitor-type weights and within-type means help explain the overall mean. The weighted-mean relationship is approximate for rounded published values; we do not run a fixed-composition scenario or a factor-contribution decomposition.

All comparisons use nominal MOP, so changes may include prices and purchase mix. No tourism price-index deflation, Shapley decomposition or quarterly spending forecast is implemented. The 16 observations support description and consistency checks, not claims about real purchase volumes, causal contributions or future expenditure.

Links to course methods

Method	Course basis	Application
Wrangling, joins, groups and pivots	Assignment 01 Module 02; Tutorial 04-01, pp. 30-42	Dates, missing values, units, sums and one-to-one joins
Descriptive statistics and empirical distributions	Lecture 04, pp. 19-56; Tutorial 07, pp. 7-18	Mean, median, IQR, P90 and ECDF
Proportions, weighted means and stratification	Lecture 04, pp. 20-26; Tutorial 05-02, pp. 3-10, 43-49	Type shares, within-group comparisons and confounding limits
Multiple OLS and Pipeline	Assignment 01 Module 04	Category encoding, trend scaling and calendar regression
Ridge and model evaluation	Modules 04/05 and Ridge page; Tutorial 10-02, pp. 7-9	Validation, MSE/RMSE/R^2 and residuals; auxiliary MAE
Matplotlib plots	Lecture 06, pp. 8-84, 91, 95; Lecture 05, pp. 37-40	Lines, distributions, heatmap, bars and error plots

Sampling distributions, confidence intervals, bootstrap and hypothesis tests are also taught. However, daily records are ordered and span different conditions, while expenditure is available only as published aggregates. We describe the finite historical periods directly rather than treating the records as independent random samples for ordinary t tests, random permutations or day-level bootstrap. Explaining this choice and its assumptions supports the course learning outcomes (University of Macau, 2026a, 2026f).

Completed outputs and research value

The work combines official data preparation, complete-date and component checks, distribution and stratified comparisons, cross-frequency reconciliation, and chronological evaluation of four calendar models. Two official Macao data sources and taught methods support a coherent investigation. Detailed records and source locations make checks possible; they do not establish independence or validity by themselves.

Deliverables include two notebooks with 16 executed code cells and 13 figures, Python scripts, fixed input data, result tables and source information. Both notebooks explain each step, its course basis, results and relevance to the proposal in English. A Dash application is not implemented or needed to reproduce the analysis.

The results include limited model fit, larger errors on high-arrival days, and changes in the direction of spending comparisons across benchmarks. Reporting these findings is part of the analysis, not a reason to omit them. The contribution is to show distribution and composition differences behind aggregate growth and evaluate the limits of the taught models.

Success criteria

Questions and evidence. Every RQ has a corresponding table or figure. Claims identify the comparison period, observation unit, denominator and nominal price basis. Unimplemented methods and unsupported interpretations are explicit.

Data and traceability. Dates and keys pass checks; source components reconcile; missing values, type differences and rounding differences remain documented. Complete daily records join to 16 distinct quarters without manufacturing daily spending observations.

Models and evaluation. Preprocessing is fitted only on training windows; tuning and selection use 2024 validation. All four test results, negative R^2 values and high-arrival errors are reported. Success does not depend on an arbitrary high R^2 threshold: a well-supported finding of limited performance is valid.

Charts and reproducibility. Axes, units, baselines, legends and sources are clear, with comparable scales for the same measure. Notebooks run in order without errors, and reported numbers match tables at their displayed precision. Data, scripts and outputs remain available; explanations are rechecked after any data update.

Main findings

RQ1 Total recovery and changing composition

Total arrivals increased by 663,179, or 1.68%, from 2019 to 2025. Overnight arrivals fell and same-day arrivals rose. The overnight share declined from 47.28% to 41.29%, a fall of 5.99 percentage points. This is a change in aggregate composition, not a longitudinal finding about individual visitors.

Measure	2019	2025
Total arrivals	39,406,181	40,069,360
Overnight arrivals	18,632,699	16,544,417
Same-day arrivals	20,773,482	23,524,943
Overnight share	47.28%	41.29%

Mainland China increased by 1,093,945 arrivals and its share rose from 70.86% to 72.42%. The Republic of Korea, Japan and Hong Kong declined by 195,388, 136,328 and 53,512 arrivals, respectively. The Mainland increase equals 164.95% of the overall net gain because other declines offset it. This is an arithmetic contribution to net change, not a visitor share or causal effect.

Excluding flagged type dates changes the 2019 overnight share by only -0.0114 percentage points and does not change 2025. This sensitivity comparison removes whole dates, not just their one-arrival discrepancies. The primary result retains all dates.

RQ2 Typical days and upper arrivals recover differently

Both years contain 365 days. In 2025, the mean was slightly higher but the median slightly lower. IQR and P90 were higher, while the maximum was lower. Total growth does not mean that all dates recovered together or that every peak increased.

Daily statistic	2019	2025
Mean	107,962.14	109,779.07
Median	104,695	103,252
IQR	27,163	34,032
P90	138,130.0	148,850.2
Maximum	226,326	221,857

Changes also differ by calendar stratum. August daily means rose from 116,874.71 to 136,097.87, while February means fell from 126,632.18 to 112,399.43. Saturday medians rose from 129,615 to 138,702; Tuesday medians fell from 93,965 to 92,336.5. Moving holidays and other conditions still differ within these strata, so the comparisons remain descriptive.

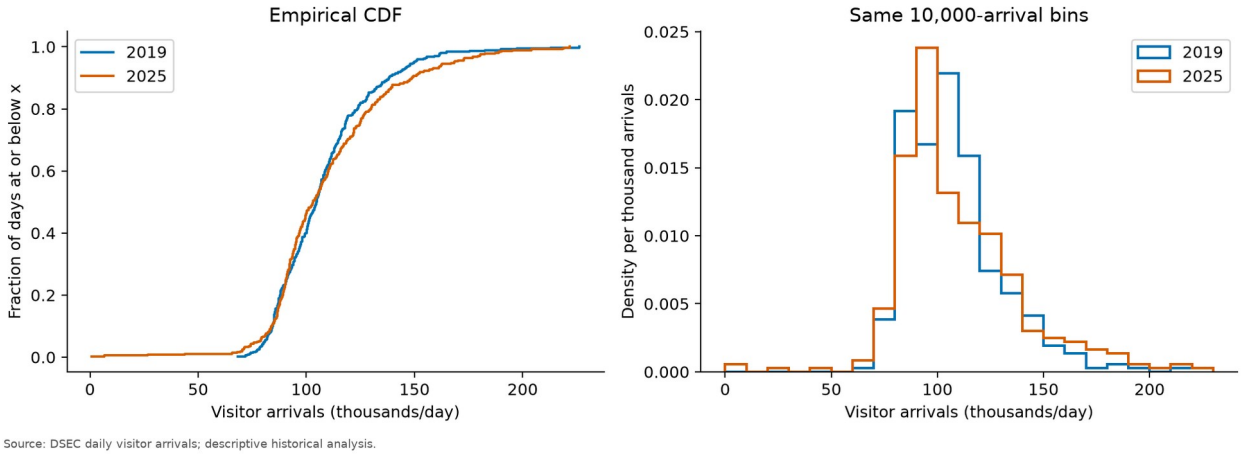


Figure 1. Daily arrival distributions. The horizontal axes show thousands of arrivals per day. The left panel shows the proportion of dates at or below each value; the right uses density with common bin widths. Source: DSEC; generated by analysis.ipynb.

RQ3 Model performance depends on the evaluation period

The four 2024 validation folds select Ridge with alpha = 100. The table retains all candidates. Validation RMSE is the square root of mean fold MSE. RMSE and MAE are in arrivals per day; R^2 is unitless.

Model	Validation RMSE	Test RMSE	Test MAE	Test R^2
Weekday mean	26,133.08	33,832.00	25,585.71	-0.3491
OLS calendar	30,876.38	33,218.97	25,113.43	-0.3007
OLS calendar + trend	73,753.37	23,916.19	16,728.15	0.3258
Ridge calendar + trend	24,820.82	25,818.77	18,767.15	0.2143

Ridge performs best in validation, but OLS with a trend has lower 2025 test error. Ridge therefore does not dominate every comparison. The high mean validation error for OLS with a trend is driven mainly by

its first fold. Rankings change with the training history and evaluation period. We retain the validation-selected Ridge and report the test comparison without using it to retune the model.

Ridge test R^2 is 0.2143: its squared error is about 21.43% lower than the reference based on the observed test mean. This is not prediction accuracy. Calendar variables leave much daily variation unexplained, so the results do not support precise peak alerts.

Actual 2025 arrivals	Days	RMSE	Mean residual
Above training P90	103	36,510.33	+27,053.09
At or below training P90	262	20,115.40	-14,061.04

The training P90 is 121,707 arrivals. Dates above it are underestimated by about 27,053 arrivals on average; other dates are overestimated by about 14,061. Opposing errors partly cancel in the overall mean. The groups use actual 2025 arrivals and are retrospective diagnostics, not a prospective classifier or an official capacity rule.

RQ4 Recovery depends on the measure and benchmark

With the corresponding 2019 quarter set to 100, 2025 first-half arrivals remain below baseline while nominal expenditure and official spending per visit are above it. All three exceed baseline in the second half, by different amounts. An index of 115.9 means 15.9% above baseline, not 115.9% growth.

2025 quarter	Arrivals index	Nominal spending index	Official per-visit index
Q1	95.2	115.9	121.7
Q2	94.3	116.1	123.2
Q3	105.4	134.1	127.3
Q4	113.0	134.7	119.2

Official annual non-gaming expenditure in 2025 is MOP 80,122 million, and official spending per visit is MOP 2,000. Annual comparisons below use official annual values. Calculations retain stored precision, while displayed figures are rounded.

Measure	2025 vs 2019	2025 vs 2024
Arrivals	+1.68%	+14.72%
Nominal non-gaming expenditure	+25.04%	+6.32%
Official nominal spending per visit	+23.00%	-7.28%

Above the pre-pandemic benchmark and below the previous year can both describe spending per visit. The table uses official annual expenditure. Using sums of quarterly expenditure instead gives 2025 growth of 25.0382% against 2019 and 6.3232% against 2024, close to but distinct from 25.0402% and 6.3218% using official annual totals. Both definitions and the rounding differences are retained.

Overnight spending per visit exceeds same-day spending in all 16 quarters. In 2025 Q3, overnight visitors account for 41.68% of arrivals but 82.22% of expenditure; official spending per visit is MOP 3,846 versus MOP 595. The annual overnight arrival share falls from 45.93% in 2024 to 41.29% in 2025. Annual overnight spending per visit falls from MOP 3,884 to 3,827, while same-day spending rises from MOP 691 to 715. An overall decline does not require both type means to fall: weights and within-type means matter. We do not estimate their individual percentage contributions.

Limitations

Observation units and dependence. Daily coverage is complete for the selected historical period, but adjacent dates may be related and conditions differ across periods. Source records share dates and summation constraints; city-quarter expenditure is not respondent microdata. Record counts do not replace an independent sample size or justify ordinary iid significance calculations.

Survey and price basis. Expenditure is a quota-survey estimate; arrivals are administrative counts joined only by quarter. Without respondent records and survey variance information, we cannot analyse personal spending distributions or construct respondent confidence intervals. Nominal amounts are not deflated and do not measure changes in real purchasing power, tourism GDP, profit or welfare.

Definitions and quality. Missing type values, one-arrival discrepancies and million-MOP rounding differences remain visible. The wording of overnight definitions differs across historical publications, which should be disclosed without inferring a statistical break from wording alone. Arithmetic agreement cannot establish that official data have no measurement error.

Selection and extrapolation. The descriptive analysis viewed 2025, and alpha tuning and candidate comparisons share the 2024 folds. The design is neither fully blind nor nested. Models contain only additive month, weekday and linear trend terms; they omit holidays, weather, events, policy and transport supply. Residual autocorrelation has not been formally tested. Error peaks cannot automatically be attributed to particular events.

Interpretation and use. Source contributions are arithmetic changes; type means and stratified comparisons are associations. There is no random assignment, causal identification, fixed-composition counterfactual or Shapley decomposition. Publication latency is unknown, and historical 2025 performance does not establish future or real-time reliability.

Files and reproducibility

analysis.ipynb covers daily distributions, annual and stratified tables, source changes, validation, test metrics and residuals. spending_analysis.ipynb covers quarterly recovery indices, visitor-type shares, official-versus-derived spending per visit and annual comparisons against both benchmarks.

The data folder contains the fixed inputs and measurement notes; results contains machine-readable tables; figures contains 13 plots. Python scripts and requirements.txt support reruns, and protocol.txt records the design. Official source URLs are retained in the data and documentation. The English edition retains the data, numerical methods and model specification; explanations must be rechecked whenever inputs or code change.

References

OpenAI. (2026). *Codex* [Generative AI tool]. https://openai.com/codex/

Statistics and Census Service. (n.d.-a). *Tourism statistics database* [Database]. Retrieved October 28, 2026, from https://www.dsec.gov.mo/TourismDBWeb/

Statistics and Census Service. (n.d.-b). *Visitor expenditure survey* [Statistics]. Retrieved October 28, 2026, from https://www.dsec.gov.mo/en-US/Statistic?id=404

Statistics and Census Service. (2017, September 29). *DSEC launches tourism statistics database* [Press release; English translation of title]. Office of the Secretary for Economy and Finance. https://www.gsef.gov.mo/zh/posts/1475

University of Macau. (2026a). *CISC7204 course intended learning outcomes* [Course materials]. UMMoodle. https://ummoodle.um.edu.mo/mod/page/view.php?id=1149854

University of Macau. (2026b). *CISC7204 course use of AI* [Course policy]. UMMoodle. https://ummoodle.um.edu.mo/mod/page/view.php?id=1149869

University of Macau. (2026c). *CISC7204 Lecture 04 exploratory data analysis* [Lecture slides]. UMMoodle. https://ummoodle.um.edu.mo/mod/resource/view.php?id=1208276

University of Macau. (2026d). *CISC7204 Lecture 05 introduction to data visualization plots* [Lecture slides]. UMMoodle. https://ummoodle.um.edu.mo/mod/resource/view.php?id=1218824

University of Macau. (2026e). *CISC7204 Lecture 06 introduction to Matplotlib* [Lecture slides]. UMMoodle. https://ummoodle.um.edu.mo/mod/resource/view.php?id=1231451

University of Macau. (2026f). *CISC7204 Lecture 10 data and sampling distributions* [Lecture slides]. UMMoodle. https://ummoodle.um.edu.mo/mod/resource/view.php?id=1266908

University of Macau. (2026g). *CISC7204 Tutorial 05-02 exploring relationships between two variables* [Lecture slides]. UMMoodle. https://ummoodle.um.edu.mo/mod/resource/view.php?id=1218848

University of Macau. (2026h). *CISC7204 Tutorial 07-01 CDFs* [Lecture slides]. UMMoodle. https://ummoodle.um.edu.mo/mod/resource/view.php?id=1241363

University of Macau. (2026i). *CISC7204 Tutorial 10-02 estimation* [Lecture slides]. UMMoodle. https://ummoodle.um.edu.mo/mod/resource/view.php?id=1266953

University of Macau. (2026j). *Module 02 data wrangling* [Course materials]. UMMoodle. https://ummoodle.um.edu.mo/mod/page/view.php?id=1153067

University of Macau. (2026k). *Module 04 model development* [Course materials]. UMMoodle. https://ummoodle.um.edu.mo/mod/page/view.php?id=1153103

University of Macau. (2026l). *Module 05 introduction to Ridge regression* [Course materials]. UMMoodle. https://ummoodle.um.edu.mo/mod/page/view.php?id=1153124

University of Macau. (2026m). *Module 05 model evaluation and refinement* [Course materials]. UMMoodle. https://ummoodle.um.edu.mo/mod/page/view.php?id=1153121